

## Wrapper Classes

- Har **primitive** datatype ki ek **wrapper class** hoti ha, or vo class **non-primitive** hoti ha.  
Example: int = integer class, short = short class, float= float class.

## Autoboxing / Unboxing

### Autoboxing:

- It automatically change “**primitive datatype**” to their respective WRAPPER CLASSES “**non\_primitive datatype**”.  
**Example:** int x = 10; —> Integer y = x;
- Internally kucha asa hota ha—> **Integer y = new Integer (x);** // old java internally perform this.  
**Integer y = Integer.value of (x);** // new java internally perform this.

### Unboxing:

- It automatically change “**Wrapper classes** to their respective **primitive datatypes**.”
  - **Example:** Integer x = 10; —> int y= x;
  - Internally kuch assa hota ha.---> **int y = x.intValue();**
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- **Autoboxing & unboxing is done with Assignments.**
  - **Autoboxing & unboxing is done with Methods calls.**
  - **Autoboxing & unboxing is done with Arithmetic operations.**
  - **Primitive can't store “null”.So we can't store wrapped classes default null in primitive datatype.**
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**Que) Why Integer x= 200; / Integer y = 200; ---> System.out.println(x==y); ---> false.**

**Ans) “==”**primitive datatype ke variables ke liye **values** ko compare krta ha, pr uper wale case me “==”  
Wrapped class ki values ke refrence ko compare kr ra ha jo stack memory me ha. Or refrence ke ander value ka address store hota ha. To “==” x,y ke value ke address ko compare kr rha ha, na ki value ko.

```
public class Autoboxing {
    public static void main(String[] args) {
        int x = 100;
        int y = 100;
        System.out.println(x==y); // Ans--> true.

        Integer a = 200;
        Integer b = 200;
        System.out.println(a==b); // Ans--> false. (hum values ko nahi balki x,y refrence
ke address ko compare kr rhe ha).
        // Agar muj values ko compare krna ha to muj asa krna hoga-->
        System.out.println(a.intValue()== b.intValue()); // true.
        // or me ya bhi krr skta hu.
        System.out.println(a.equals(b)); // true.

        Integer p = 100;
        Integer q = 100;
        System.out.println(p==q); // Ans----> true. (it use caching)
        // matlab (int -128-----127) tk ki values ke liye fix refrence hota ha,
        // to jab hum call krte ha to direct unki values call hoti ha.
    }
}
```

### **POJO CLASS “plain old java object”**

- Isme koi hardcore business logic nahi hona chahiye.
- Isme sirf getter/setter, constructor, variables(field).